

ROBERT PAULS

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PROFILE

Computer science graduate specialising in software engineering and AI/ML, with hands-on experience shipping full-stack apps (React/TypeScript, Flask, GCP/Firebase) and computer vision models (PyTorch, OpenCV, YOLO) from concept to cloud deployment. Independently designed and deployed cloud infrastructure, debugged cross-stack issues end-to-end, and built ML pipelines from scratch rather than relying on off-the-shelf tooling. Pairs this with three-plus years of hardware diagnostics, bringing rare end-to-end technical grounding to graduate software engineering or AI/ML roles.

EXPERIENCE

Computer Repairs, Hornchurch - Technician

Aug 2022 – Present

- Diagnose and repair desktop systems, cutting average turnaround time by 30% through systematic hardware/software fault-finding.
- Advise customers on tailored hardware builds for gaming, creative, and business use cases, translating technical trade-offs into plain-language recommendations.

Rok Co., Romford - Assistant

Feb 2015 – Mar 2021

- Supported a small business by completing 100+ handyman tasks, maintaining repeat custom and resolving client issues on first visit in most cases.

EDUCATION

City St George's, University of London - BSc Computer Science (2:1)

Sept 2021 – June 2026

All Saints Catholic School, Dagenham

Sept 2014 – May 2021

- A-Levels: Design and Technology (A), Computer Science (B), CTEC Business Studies (Pass).
- GCSE: 10 GCSEs (9-4) including Computer Science.

TECHNICAL SKILLS

Programming/Scripting: CSS, HTML, JavaScript, Python, SQL, TypeScript.

Frameworks & Libraries: Flask, Node.js, OpenCV, Pandas, PyTorch, React, TensorFlow, Vite, YOLO.

Databases & Cloud: Cloud Run, Firebase, Firestore, GCS, GitHub Actions, MySQL, MariaDB.

Tools & Platforms: GitHub, Linux, Roboflow, Visual Paradigm, VirtualBox, Windows.

Hardware: Assembly, Maintenance, Optimisation, Troubleshooting.

PROJECTS

- [F1StatFinder](#): full-stack F1 statistics platform with automated data pipelines, built solo end-to-end across frontend, backend, and cloud infrastructure.
 - Built a Firestore caching layer with decade-bucketed document structures in front of the FastF1 and Ergast APIs, working within Firestore's document size limits.
 - Resolved cross-stack deployment issues - build configuration, IAM permissions, environment variables - while learning new cloud tooling.
 - Automated weekly data sync via Cloud Scheduler, serving a React/TypeScript frontend on Firebase Hosting.
 - Built an LLM-powered race summary feature via prompt engineering against a production API, and used Claude throughout development to speed up debugging and architecture decisions.
- [F1Detect](#): real-time object detection pipeline for identifying F1 cars in footage.
 - Built custom OpenCV scripts to extract and sample frames from raw footage, then trained YOLOv11 with CUDA acceleration, resolving training-stability issues through iterative debugging.
 - Learned Roboflow and ran an evolutionary hyperparameter search on a custom dataset blending video game and real-world footage, with a benchmarking harness for per-frame timing.
 - Achieved ~60 FPS real-time inference with 0.75 mean detection confidence across 1,100+ frames.
- [Face Cover Detection](#): multiclass mask detection system comparing computer vision pipelines.
 - Labelled a personal test set and cleaned/preprocessed the coursework dataset, addressing class imbalance across 2,800+ images.
 - Implemented HOG, SIFT/Bag-of-Visual-Words, and CNN pipelines from scratch, tuning each via grid search and stress-testing against synthetic noise, rotation, and affine warping.
 - Achieved the best accuracy (91%) with a custom-built CNN model.
- [Phishing Email Detection](#): NLP-based classifier distinguishing phishing emails from legitimate ones.
 - Cleaned and tokenised email text, removing nulls, duplicates, and empty bodies.
 - Built a custom CBOW Word2Vec model from scratch in PyTorch, evaluating embeddings against bag-of-words baselines (DT, SVC, NB).
 - Achieved 96% accuracy with an MLP on CBOW embeddings, outperforming Random Forest's 93%.

HOBBIES AND INTERESTS

- Motorsports: following Formula 1 and other motorsport categories closely.
- Hardware research: keeping on top of the latest desktop and server component advancements.
- Computer systems: self-taught in hardware assembly, maintenance, and troubleshooting.